



Sparkfun Triad Spectroscopy Sensor

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Guess to Win!

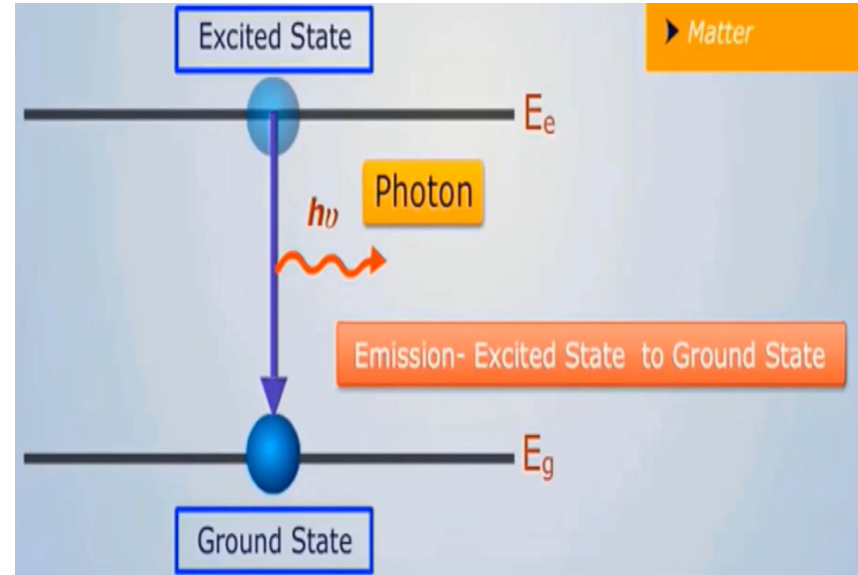


Who can afford it?



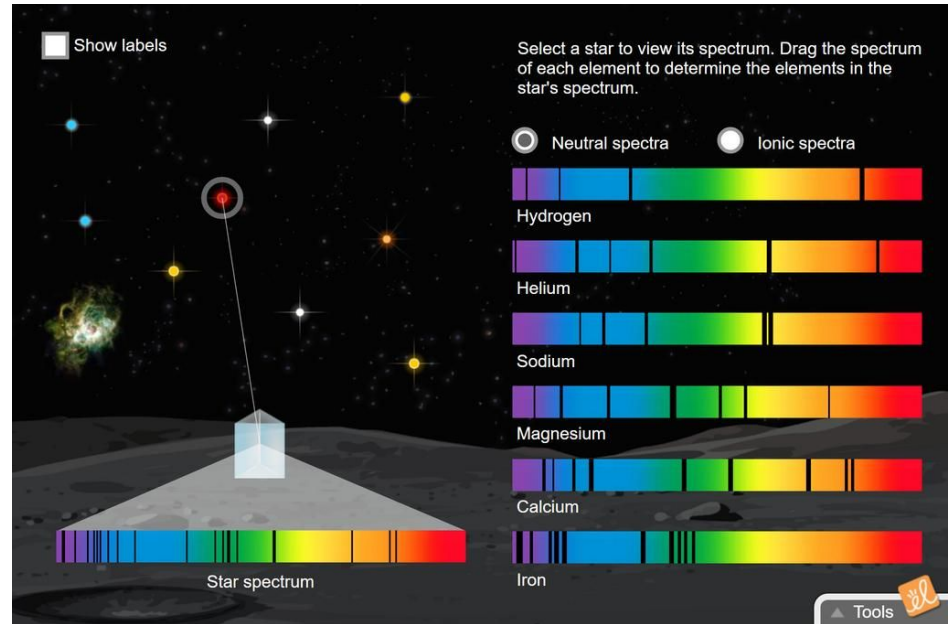
The Physics Pt.1

- Electron absorbs energy -> excited state -> releases photon -> lower state
- Low energy -> large wavelength
- High energy -> small wavelength

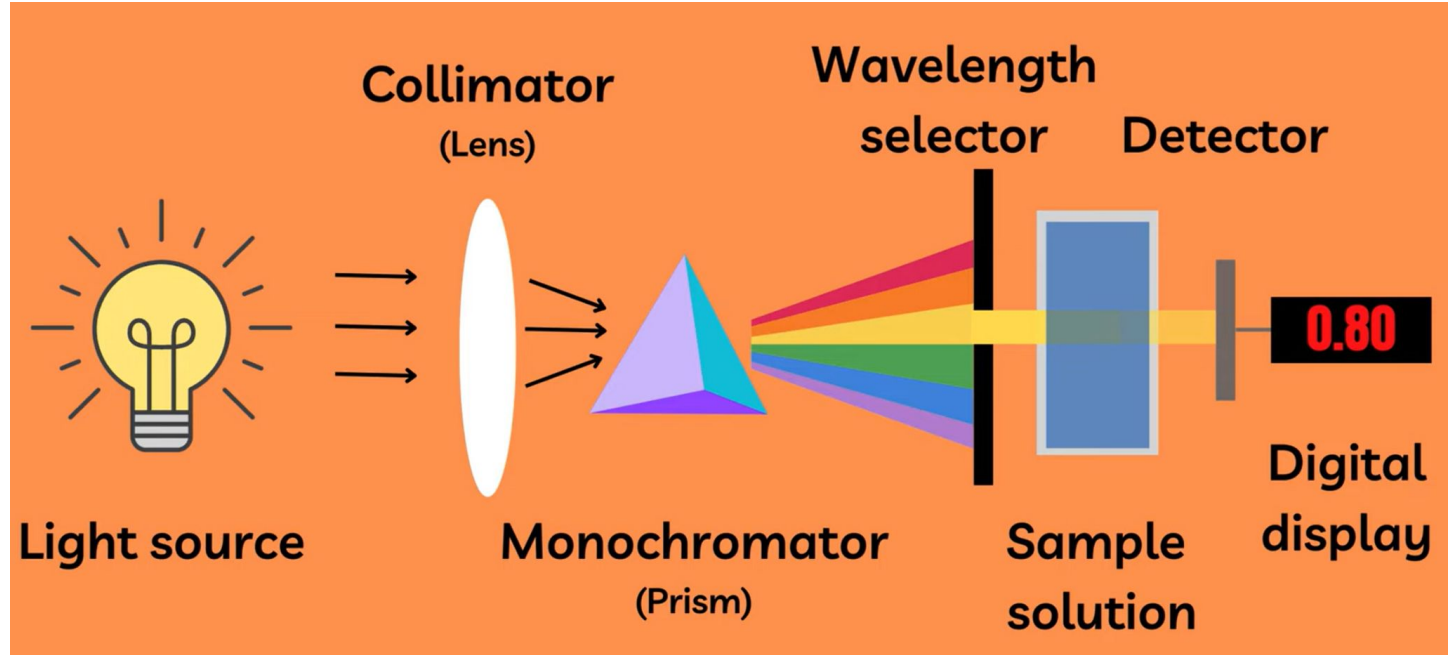


The Physics Pt.2

- Different spectrums = different excitement = different chemicals
- Any object reflecting light is a source
- Separate into different light branches w/ slit, filter with optical filter
- Intensity of light intensity = value



Diagram



AS7265x's relatives



AS7331 (\$28)

3 Channels

UVA,UVB,UVC



AS7262 (\$28)

6 Channels

Visible (430-670nm)

40nm FWHM

Chlorophyll



AS7263 (\$30)

6 Channels

NIR (610-860nm)

20nm FWHM

Useful for NDVI



AS7343 (\$22)

14 Channels

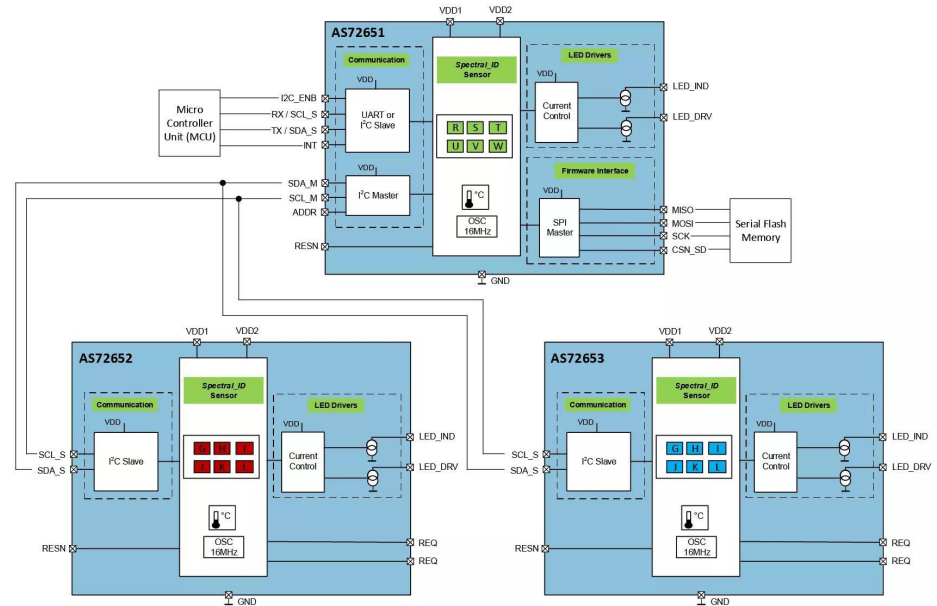
11 visible, 1 NIR, Flicker
,Clear

20nm FWHM

No calibration!

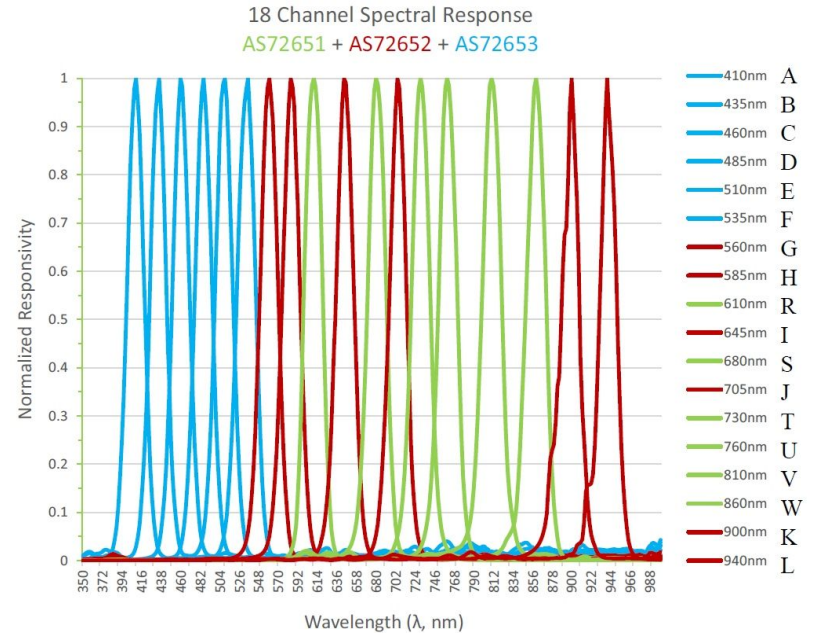
AS7265x

- Three sensors (AS72651, AS72652, AS72653)
- Infrared (51), Visible (52), UV (53)
- Master (51) -> Slaves (52,53)
- Temperature Compensation
- I2C / UART interface
- SPI Master -> Serial Flash Memory (firmware updates)



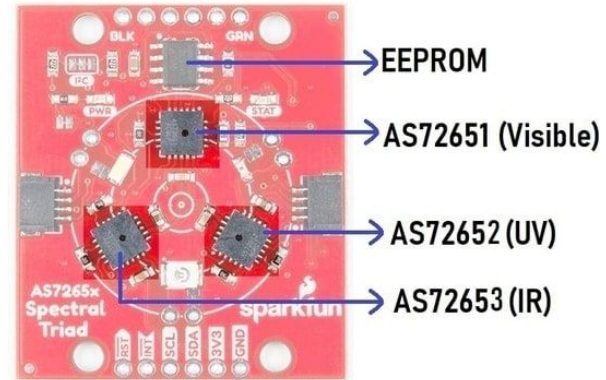
Spectroscopy specs

- 18 bands (410nm-940nm), 6 per die
- 20nm spectral resolution
- 28.6 nW/cm² per count
- Not a true mass spectrometer
 - Detects chemical composition well
 - But inaccurately determines concentration



Sparkfun Triad Spectroscopy Sensor

- Three incandescent bulb through holes
- Qwiic I2C plug & play
- Visible, infrared, and UV LED illumination
- AMS Firmware EEPROM Flash Chip
- Factory calibrated
- Only **\$70**



Application – NDVI (6 band AS7263)



AS7265x Applications - Full Tricorder



Questions?